



PRODUCT BIBLE · POC EDITION · V3.0 · CONFIDENTIAL

Working Demo in 7 Weeks. First Customer by Week 13.

A scoped, honest, achievable Proof of Concept — CO₂ storage simulation in the browser with real formation presets, project save/load, geomechanical inputs, LAS file import, and a jurisdiction toggle. No over-promising. No scope creep. Ship and validate.

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What This PoC Is (and Is Not)

NAMING DECISION – REQUIRED BEFORE LAUNCH

The name "CarbonLens" implies full CCUS coverage: Capture, Utilization, *and* Storage. The PoC covers only the storage component. Launching with a "full CCUS" positioning and delivering a storage-only tool will immediately erode credibility with any industry professional or higher institution reviewer. **The PoC must launch as "CarbonLens Storage Studio"** – explicit about scope. The full CCUS platform comes later and is documented separately. This is not a compromise; it is what earns trust.

Honest Scope Statement

✓ IN SCOPE – POC

- ✓ Parametric CO₂ storage reservoir builder (3D + 2D)
- ✓ Real-time CO₂ plume simulation (Web Worker engine)
- ✓ ML property engine (IFT, density, viscosity, solubility, phase)
- ✓ P10/P50/P90 uncertainty envelopes
- ✓ Project save/load (JSON export + Cloudflare KV)
- ✓ Real formation presets (Sleipner, Decatur, Otway)
- ✓ LAS 2.0 file import (auto-populate formation params)
- ✓ Geomechanical tab (FPG, Eaton correlation, MAIP)
- ✓ Jurisdiction toggle (EPA / UK NSTA / NPD / PETRONAS)
- ✓ Basic permit table export (PDF, EPA Class VI template)
- ✓ Guided onboarding tutorial ("First simulation in 3 minutes")

✗ DEFERRED TO FULL PRODUCT

- ✗ Capture module (source characterisation, technology selector)
- ✗ Transport module (pipeline sizing, pressure drop)
- ✗ Utilization module (CO₂-EOR MMP, mineralization)
- ✗ Full MMV planner (monitoring well locations, schedules)
- ✗ Team collaboration and version history
- ✗ Monte Carlo n=1,000 batch engine
- ✗ GIS shapefile import
- ✗ TOUGH2/CMG export formatters
- ✗ Enterprise security (SOC 2, SSO, audit logs)
- ✗ Full academic partnership programme
- ✗ Subsurface AI assistant

What the PoC Proves

The PoC validates three hypotheses that the full production investment depends on:

- **Hypothesis 1 — Visual Demo Closes Sales:** A CCUS engineer who sees the 3D reservoir builder with a real Sleipner preset running will request a paid account before the demo ends. Validated when: 3 paying customers by Week 13.
- **Hypothesis 2 — Client-Side ML Is Fast Enough:** Extracted TypeScript equations update property values and simulation in real time with no perceptible lag, even on mid-range hardware. Validated when: <50ms property update confirmed on a 2021 MacBook Air M1 and a Windows i5 laptop.
- **Hypothesis 3 — Academic Adoption via Presets:** A professor can load the Sleipner case in one click and run a simulation that matches published P50 plume extent within 15%. Validated when: UTP faculty or one international collaborator uses it in a research context within 60 days of launch.

PoC Feature Set & Priority Order

PRIORITY LOGIC

P0 = deal-breakers. The PoC cannot go live without these. P1 = professional credibility — must be in before the first paying professional customer. P2 = differentiation — adds in the first 60 days post-launch. Anything not in this list is documented in the Full Production Bible.

P0 — MUST HAVE AT LAUNCH

Project Save / Load

A professional who closes their browser tab must not lose their 6-hour reservoir model. JSON export on free tier. Cloudflare KV cloud save on paid tiers. This was the most critical omission in v2.

P0 — MUST HAVE AT LAUNCH

Real Formation Presets

Sleipner, Decatur (Illinois Basin), and Otway Basin as one-click validated presets. Academic trust and demo credibility depend on these. A professor who can load Sleipner and reproduce published results adopts the tool immediately.

P1 — BEFORE FIRST PAYING PRO CUSTOMER

LAS 2.0 File Import

Every professional site characterisation has a well log in LAS format. Without this, the parametric builder is a teaching toy. With it, real engineers can load their actual site data. ~200 lines of JavaScript.

P1 — BEFORE FIRST PAYING PRO CUSTOMER

Geomechanical Tab

EPA Class VI requires mechanical integrity analysis: fracture pressure gradient (FPG), Eaton minimum horizontal stress, MAIP calculation. Without this, the permit suite is incomplete and cannot claim to auto-populate Class VI templates credibly.

P1 — BEFORE FIRST PAYING PRO CUSTOMER

Jurisdiction Toggle

EPA Class VI / UK NSTA / Norwegian NPD / PETRONAS Malaysia. Unlocks the non-US market immediately. The PETRONAS/Malaysia jurisdiction is a first-mover opportunity uniquely accessible from UTP. Output templates switch with the toggle.

P1 — BEFORE FIRST PAYING PRO CUSTOMER

Guided Onboarding Tutorial

“Build your first CO₂ storage simulation in 3 minutes.” Pre-loaded sample formation, step-by-step overlay, inline parameter documentation linking to source papers. Critical for academic adoption and reducing time-to-first-value.

P2 — FIRST 60 DAYS POST-LAUNCH

Basic MMV Planner

Monitoring well location calculator (distance from injection well based on AoR P90), pressure monitoring schedule generator, USDW protection zone delineation. Completes the permit suite.

Real Formation Presets

WHY PRESETS ARE PO

A professor at UT Austin or UTP can say to their class: “Load the Sleipner case, run the simulation, compare your P50 plume extent to the result in Celia et al. (2015).” This single capability transforms CarbonLens from a demo tool into a legitimate research and teaching platform. It is also the most powerful sales demo available — showing a real formation, not a toy example.

The Three Launch Presets

FORMATION	LOCATION	WHY SELECTED	KEY PARAMETERS	VALIDATION REFERENCE
Sleipner (Utsira Sand)	North Sea, Norway	World’s most-studied CO ₂ storage site. 30 years of public data. Every CCUS researcher knows it. Instant credibility.	Depth: 800m, T: 37°C, P: 10.5 MPa, φ: 35–40%, k: 1–8 D, NaCl: 1.2 mol/kg	Celia et al. 2015; Arts et al. 2004 (seismic monitoring)
Illinois Basin — Decatur	Illinois, USA	Well-characterised US saline aquifer. Most-cited US Class VI case. Strong for EPA-jurisdiction demos with consultancy ICPs.	Depth: 1,800m, T: 54°C, P: 17.8 MPa, φ: 10–15%, k: 50–300 mD, NaCl: 2.0 mol/kg	Finley (2014) MGSC project; EPA permit data
Otway Basin	Victoria, Australia	CO ₂ CRC public dataset. Unlocks Asia-Pacific market demo. Important for academic users outside US/EU.	Depth: 2,050m, T: 80°C, P: 20.8 MPa, φ: 12–18%, k: 100–500 mD, mixed brine	CO ₂ CRC Otway Project final reports (public)

Implementation

Each preset is a JSON object stored in the application bundle containing: all reservoir parameters, geological description, source citation, expected P50 plume extent at 10/20/50 years (from published literature), and a formation thumbnail image. Users click “Load Preset” from the formation panel — the reservoir builder populates all inputs and renders the formation immediately. A citation badge appears in the UI referencing the source publication.

Future Presets (Post-PoC)

Weyburn-Midale (Canada), Quest (Alberta), Quest CarbonSAFE (US Midwest), In Salah (Algeria), and a Malaysia Offshore formation (PETRONAS CCS pipeline). The PETRONAS preset is strategically important — it makes CarbonLens the first web-based tool with a Malaysia-specific CO₂ storage case, which is a first-mover advantage accessible from UTP.

Persistence, LAS Import & Geomechanics

04 — Project Persistence

A professional consultant working on a Class VI application may spend 6 hours building and calibrating a reservoir model. Losing that work when the browser tab closes is a deal-breaker for every professional ICP. This is the most critical omission from v2. It must be in the PoC at launch.

TIER	SAVE METHOD	IMPLEMENTATION
Free (Explorer)	JSON export / import only	Single “Save Project” button exports all Zustand state as a .carbonlens JSON file. “Load Project” reads it back. Zero infrastructure. Client-side only. Implement in 2 hours.
Researcher / Professional	Cloudflare KV cloud save	On save, POST Zustand state JSON to a Cloudflare Worker, stored in KV keyed to Stripe customer ID. Load on login from any device. Cloudflare KV free tier: 100K reads/day, 1K writes/day — sufficient at PoC scale.
Studio (future)	Named project library + team sharing + version history	Cloudflare D1 (SQLite at edge) for project metadata. Version history as append-only KV entries. Team sharing via project UUID invitation link. Post-PoC feature.

05 — LAS 2.0 File Import

LAS (Log ASCII Standard) files contain the actual well log data that engineers use to characterise formations: depth intervals, gamma ray, resistivity, porosity, density. Without LAS import, professionals must manually re-enter data they already have in a standard format — which means the tool is a teaching aid, not a professional instrument.

Implementation: A pure JavaScript LAS 2.0 parser reads the header section (metadata, units) and data section (depth-indexed curve values). The parser extracts: TVD (depth), PHIE (porosity), PERM or estimated permeability, RHOB (bulk density), and temperature from the geothermal gradient. These auto-populate the corresponding formation panel inputs. Curves that cannot be mapped are shown in a “unmatched curves” list for manual assignment. ~200 lines of TypeScript. No external library needed — LAS 2.0 is a simple text format.

06 — Geomechanical Module

EPA Class VI applications require a mechanical integrity test plan, fracture pressure analysis, and the MAIP calculation. Without a geomechanical module, the permit suite literally cannot complete a Class VI application.

CALCULATION	METHOD	OUTPUT
Overburden Stress (σ_v)	Integrated bulk density from LAS or estimated from depth gradient (22.6 MPa/km default)	Vertical stress at injection depth (MPa)
Min. Horizontal Stress (σ_h)	Eaton (1969) correlation: $\sigma_h = [v/(1-v)] \times (\sigma_v - \alpha P_p) + \alpha P_p$	Minimum horizontal stress (MPa) — fracture propagation threshold
Fracture Pressure Gradient (FPG)	FPG = σ_h / depth (MPa/m) — displayed as psi/ft for US jurisdiction	FPG (MPa/m and psi/ft) with depth profile chart
MAIP	MAIP = FPG × injection depth × safety factor (0.9 default, EPA-recommended)	Maximum Allowable Injection Pressure (MPa) — permit-ready value
Caprock Seal Index	CO ₂ column height = $(2 \gamma \cos\theta) / (r \rho g)$ from IFT Model 1 output	Maximum storable CO ₂ column height (m) — linked to ML IFT output

GEOMECHANICS LINKS TO ML ENGINE

The caprock seal index directly uses the IFT output from Model 1 — the ML engine feeds the geomechanical calculation. As IFT changes with T/P/salinity parameters, the seal index updates live. This integration is a genuine differentiator: no other web tool links ML-predicted IFT to caprock seal capacity in real time.

Jurisdiction Toggle & IP Strategy

07 — Regulatory Jurisdiction Toggle

The v2 product is entirely EPA/45Q-centric. But the risks section correctly identified that 60% of the global CCUS pipeline is outside the US. A jurisdiction toggle is the lowest-effort, highest-impact feature for unlocking non-US markets — and the PETRONAS/Malaysia jurisdiction gives CarbonLens a uniquely accessible first-mover position from UTP.

JURISDICTION	REGULATOR	KEY DIFFERENCES FROM EPA	MARKET OPPORTUNITY
USA — EPA Class VI	US EPA Underground Injection Control	Baseline. MAIP, AoR, MVA plan required. 45Q tax credit driver.	232+ active permit applications. Primary market.
UK — NSTA	North Sea Transition Authority	Storage permit (SP) + development consent. No AoR equivalent — uses plume footprint. CO ₂ storage licensing regime.	Drax BECCS, BP Net Zero Teesside, HyNet. £billions committed.
Norway — NPD	Norwegian Petroleum Directorate	Petroleum Act covers CO ₂ storage. Sleipner/Northern Lights standards. Detailed geomechanical requirements.	Equinor Northern Lights — world’s first commercial CCS. Academic benchmark market.
Malaysia — PETRONAS	PETRONAS & Ministry of Natural Resources	No standardised framework yet — PETRONAS-guided. Offshore saline aquifer focus. EOR CO ₂ projects.	First-mover opportunity. No web-based permitting tool exists. UTP connection gives direct access to PETRONAS network.

Implementation: A “Regulatory Jurisdiction” dropdown in the site screening panel switches: (1) output template structure for permit export, (2) calculation nomenclature (AoR vs. plume footprint), (3) pressure unit display (MPa vs. psi), (4) depth references (TVDss vs. KB). The underlying physics does not change — only the output formatting and labelling.

08 — IP Strategy: The Client-Side Contradiction, Resolved

THE PROBLEM V2 DID NOT ADDRESS

CarbonLens’s competitive moat is “ML equations running client-side as TypeScript.” But any user can open DevTools, navigate to the JavaScript bundle, and extract those equations in 5 minutes. Both product

The Committed Strategy: Hybrid Architecture

The PoC commits to a **hybrid architecture** that balances the speed advantages of client-side computation with meaningful IP protection:

- **Client-side (visible):** The classical baseline equations (Span-Wagner EOS, Duan-Sun solubility, Fenghour viscosity, Li et al. IFT correlation). These are already published in peer-reviewed literature — there is nothing to protect here, and being transparent about using them builds trust.
- **Server-side (protected):** The ensemble weights and the bootstrap uncertainty calibration that produce the P10/P50/P90 bounds. These are the trained ML outputs — the part that took months to produce and that competitors cannot replicate without the same data. A lightweight Cloudflare Worker receives T/P/salinity, returns calibrated P10/P50/P90 values. This is one API call per property update — still fast (<100ms), but the weights are never in the bundle.
- **Data moat (irreplaceable):** The proprietary UTP lab measurement dataset cannot be reverse-engineered from DevTools. This is the strongest long-term protection — more robust than any obfuscation strategy.

What this means for the architecture: The simulation engine and geomechanical calculations run entirely client-side. The ML ensemble prediction endpoint is a Cloudflare Worker that returns calibrated confidence bounds. This hybrid adds ~\$5/month to infrastructure costs and adds one Worker to the stack. Update this in both documents and in the patent application claims.

Success Criteria & Path to Full Product

09 — PoC Success Criteria

The PoC is declared successful when all of the following are met within 90 days of launch:

3 Paying customers at any tier by Week 13	1 Academic or research user validates a preset formation against published data	<50ms Property update latency on mid-range 2021 laptop hardware
4 Jurisdictions supported in the permit output toggle at launch	3 Validated real formation presets available at launch	0 Lost projects due to missing save functionality — zero tolerance

WHAT INVALIDATES THE THESIS

If 3 paying customers are not achieved by Week 13, the primary hypothesis fails. The correct response is not to build more features — it is to talk to the people who tried the demo and did not pay. Their reason tells you what to fix. Do not add features before understanding the conversion failure.

10 — Path to Full Production

The PoC validates the core product-market fit. The Full Production Platform (documented separately) adds what the PoC deliberately defers:

POC VALIDATES	FULL PRODUCT ADDS	WHY DEFERRED
Storage simulation is visually compelling and closes deals	Capture + Transport + Utilization modules completing full CCUS coverage	Storage-only is sufficient to validate commercial viability first
ML property accuracy is acceptable for professional use	Full MMV planner, advanced UQ (n=1,000), MMP predictor for CO ₂ -EOR	Basic permit suite sufficient for PoC; full completeness is post-validation
LAS import enables professional workflow integration	GIS shapefile import, multi-well datasets, subsurface AI assistant	LAS covers 80% of professional data import needs; GIS is incremental

Cloudflare KV saves projects reliably	Team collaboration, version history, enterprise SSO, SOC 2 compliance	Single-user save is sufficient for PoC; team features are Studio+ tier
4 jurisdiction templates are sufficient for initial markets	Australia (GHGSRS), Canada (Alberta Energy Regulator), Saudi CCUS	First-mover validated before expanding jurisdiction library

THE DECISION THRESHOLD

Invest in the Full Production Platform when: (1) \$10K MRR is achieved, (2) provisional patent is filed and confirmed, and (3) at least one design-partner customer has co-designed a feature they need for a real permit application. Do not build the full product on a thesis — build it on validated demand.